

Module 1: Network + Network Fundamentals and building networks

Day 2

Task 1:

TCP/IP Layer	Which packet / field to point to
Application	An HTTP or TLS packet — e.g. "Client Hello" or an HTTP GET request
Transport	Any TCP packet, e.g. "443 → 52344 [ACK] Seq=1 Ack=1 Win=64240"
Internet	Same packet — expand the "Internet Protocol Version 4" section
Network Access	Expand "Ethernet II" or "802.11" section of any packet

Task 2:

OSI Layer	TCP/IP Layer	Example
Application	Application	Opening the WhatsApp chat screen
Presentation	Application	Photo/video getting encoded and encrypted before sending
Session	Application	Staying connected during a Zoom call or WhatsApp video call
Transport	Transport	Your Zomato order confirmation arriving fully and in order (TCP)
Network	Internet	Your phone's IP address getting your BookMyShow ticket request to the right server

OSI Layer	TCP/IP Layer	Example
Data Link	Network Access	Your phone's Wi-Fi router directing data using MAC addresses
Physical	Network Access	Wi-Fi radio signals or Ethernet cable carrying the actual bits

Task 3:

Source Port — identifies which application/process on your device sent the data (e.g. your browser's ephemeral port)

Destination Port — identifies which service on the receiving server should get the data (e.g. port 443 for HTTPS)

Sequence Number — tracks the order of bytes sent, so the receiver can reassemble segments correctly

Acknowledgment Number — confirms which bytes have been successfully received so far

Flags (SYN, ACK, FIN, etc.) — control bits that manage connection setup, data acknowledgment, and connection teardown

Window Size — tells the sender how much data the receiver can accept before needing another acknowledgment

Checksum — used to detect errors/corruption in the transmitted segment

Task 4:

Application Layer Instagram app creates the request
Presentation Layer Message split into packets, port numbers added, TCP ensure order
Session Layer Source and destination IP address added for routing
Transport Layer MAC address added, converted to Wi-Fi signals, sent to router

Task 5:

Packet: TCP, "443 → 51724[ACK] Seq=26 Ack=30"

Layer: Transport layer

How I identified it: In Wireshark's packet detail pane, the "Transmission Control Protocol" section is expanded, showing Source Port 443, Destination Port 51724, and sequence/acknowledgment numbers — these are Transport-layer fields, not Application or Internet layer ones.

Real-world action it represents: This is the equivalent of your phone confirming to Instagram's server that a piece of a photo or video you're loading has been received correctly — similar to how BookMyShow's server and your phone exchange confirmations to make sure your seat selection request arrives completely and in order before your ticket is confirmed.